

Problem 1 — Regularization to Avoid Overfitting

Build a neural network for binary classification and investigate how **regularization techniques** help reduce overfitting.

Dataset

Breast Cancer Wisconsin Diagnostic Dataset.

A. Data Understanding

After loading the dataset:

- Display `shape`.
- Display `head()`.
- Display data types.
- Generate summary statistics.
- Check for missing values.
- Check the target/class distribution.
- Identify and remove unnecessary columns such as ID.
- Convert the target variable into numerical form.
- Separate `X` and `y`.

B. Preprocessing

- Perform train-test split.
- Apply feature scaling.
- Use an appropriate scaler.
- Make sure the scaler is fitted only on the training data.

C. Baseline Neural Network

Build a relatively large neural network **without regularization**.

Use:

- Binary Cross Entropy
- Suitable optimizer
- Appropriate output activation

Train for a sufficiently large number of epochs.

D. Identify Overfitting

Plot:

1. Training loss vs validation loss
2. Training accuracy vs validation accuracy

E. L2 Regularization

Add L2 regularization to the Dense layers.

Experiment with:

$$\lambda = 0$$

$$\lambda = 0.001$$

$$\lambda = 0.01$$

F. Dropout Regularization

Now experiment with:

$$\text{Dropout} = 0$$

$$\text{Dropout} = 0.2$$

$$\text{Dropout} = 0.5$$

Compare the three models.

Problem 2 — Batch Normalization

Understand how **Batch Normalization** affects neural-network training.

Use the **same Breast Cancer dataset**.

- Build a relatively large neural network **without Batch Normalization**
- Now Build a relatively large neural network **with Batch Normalization**
- Now Build a relatively large neural network **with Layer Normalization**
- Now Compare all three Models

- **Finally Experiment**

Train both models with the same:

- Learning rate
- Batch size
- Epochs
- Optimizer
- Train/test split

Problem 3 — Compare Five Optimizers

Use exactly the **same neural-network architecture and same dataset** for all optimizers.

Optimizers

Compare these five:

1. **SGD**
2. **Momentum**
3. **NAG**
4. **RMSprop**
5. **Adam**

Use the same dataset that were used for above two problems and build a Neural network and implement these all 5 optimizers on same model structure just keep changing the optimizer and observe the changes and compare all the optimizers based on the accuracy score

Finally draw a accuracy table for all above 5 optimizers for better comparison.